C21 Acids, Bases and Buffers – Knowledge Organiser

AQA A-level Chemistry • Set 1: Definitions and pH **Paper 1**

#	Question	Answer
1	Brønsted–Lowry acid	Proton donor
2	Brønsted–Lowry base	Proton acceptor
3	Conjugate acid–base pair	Species differing by one H^+
4	Strong acid	Fully dissociates
5	Weak acid	Partially dissociates
6	pH equation	$pH = -\log[H^+]$
7	$[H^+]$ from pH	$[H^+] = 10^{-pH}$
8	Ionic product of water (K_w)	$K_w = [H^+][OH^-]$
9	Value of K_w at 298 K	1.00×10^{-14} mol^2 dm^{-6}
10	$[H^+]$ of a strong base solution	$K_w \div [OH^-]$

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AQA A-level Chemistry • Set 2: Weak Acids **Paper 1**

#	Question	Answer
1	K_a expression for HA	$[H^+][A^-] \div [HA]$
2	Units of K_a	$mol\ dm^{-3}$
3	pK_a	$-\log K_a$
4	K_a from pK_a	10^{-pKa}
5	$[H^+]$ of a weak acid	$\sqrt{(K_a[HA])}$
6	Assumption 1 for weak acid pH	$[H^+] = [A^-]$
7	Assumption 2 for weak acid pH	[HA] at equilibrium = initial [HA]
8	Larger K_a means	Stronger acid
9	Effect of raising temperature on K_w	Increases
10	Pure water at any temperature	$[H^+] = [OH^-]$, so neutral

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AQA A-level Chemistry • Set 3: Buffers **Paper 1**

#	Question	Answer
1	Buffer	Resists pH change when small amounts of acid or base are added
2	Acidic buffer	Weak acid + its salt
3	Basic buffer	Weak base + its salt
4	Example of a basic buffer	NH_3 and NH_4Cl
5	$[H^+]$ of an acidic buffer	$K_a \times [HA] \div [A^-]$
6	pH when $[HA] = [A^-]$	$pH = pK_a$
7	Buffer in blood	H_2CO_3 / HCO_3^-
8	Conjugate base of CH_3COOH	CH_3COO^-
9	Conjugate acid of NH_3	NH_4^+
10	Species that removes added H^+ in an acidic buffer	A^-

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AQA A-level Chemistry • Set 4: pH Curves and Indicators **Paper 1**

#	Question	Answer
1	Equivalence point	Moles of acid and base exactly react
2	pH at equivalence: strong acid–strong base	7
3	pH at equivalence: weak acid–strong base	Above 7
4	pH at equivalence: strong acid–weak base	Below 7
5	pH at half-equivalence (weak acid)	pK_a
6	Indicator choice rule	Changes colour within the vertical section
7	Indicator for weak acid–strong base	Phenolphthalein
8	Indicator for strong acid–weak base	Methyl orange
9	Indicator for weak acid–weak base	None suitable
10	Starting pH of 0.1 mol dm^{-3} HCl	1